

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Level 2

Teaching Guide

Level: Level 2

Class: Class 7

Duration: June - January (8 months)

Period: 40 minutes

ICT Teaching Resources Portal

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Curriculum Overview

ICT Level 2 builds on the foundations established in Level 1, designed for Class 7 students in KREIS residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for deeper conceptual understanding.

Themes Covered

Chapter	Theme	Periods	Key Topics
1	Networking and Internet	12	Network types, LAN/WAN,
2	Advanced Graphics	12	Internet services, Email,
3	Multimedia	12	Image editing, Layers, Filters, GIMP
4	Data Handling	12	Audio/Video editing, Presentation
5	Programming with Scratch	12	tools
6	Digital Citizenship	12	Spreadsheets, Charts, Formulas
			Variables, Conditions, Loops
			Online safety, Cyberbullying,
			Digital footprint

Pedagogical Approach

5E Model Implementation

Phase	Duration	Activities
Engage	5-7 min	Real-world connection, regional metaphors,
Explore	10-12 min	questions
Explain	10-12 min	Hands-on activity, Driver-Navigator pairs, Lab work
Elaborate	8-10 min	Live coding, Concept introduction, C-P-A approach
Evaluate	5-8 min	Extended activity, Parson's problems, Application
		Formative check, Exit tickets, Self-assessment

International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives to Visual blocks to Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator:** Pair programming for lab sessions
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

Lesson Plan Structure (40 Minutes)

Time	Phase	Teacher Activity	Student Activity
0-5 min	Engage	Present real-world scenario using regional metaphor	Discuss, share observations
5-15 min	Explore	Guide hands-on exploration in pairs	Work in Driver-Navigator pairs, explore software
15-25 min	Explain	Live coding demonstration, introduce concepts	Follow along, ask questions, take notes
25-35 min	Elaborate	Assign extension activity or Parson's problem	Apply learning, solve problems independently
35-40 min	Evaluate	Conduct formative assessment, assign homework	Complete exit ticket, receive dormitory task

Assessment Framework

Assessment Components (Per Class)

Assessment Type	Count	Marks Each	Total
Formative (FA-1 to FA-4)	4	30	120
Unit Tests	4	10	40
Summative (SA-1, SA-2)	2	40	80
Practical Lab Exams	2	20	40
Grand Total	12		280

Resources Required

Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)

- Projector for demonstrations
- Speakers and headphones

Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Inkscape
- Audio: Audacity
- Video: Kdenlive
- Programming: Scratch
- Browser: Firefox / Chromium

Dormitory Tasks (Zero Lab Access)

For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Network Diagrams:** Draw and label network topologies on paper
- **Spreadsheet Planning:** Design spreadsheet layouts with formulas on paper
- **Binary Conversion:** Practice decimal to binary conversions
- **Cyberbullying Scenarios:** Discuss and write responses to online safety scenarios
- **Digital Footprint Journal:** Reflect on personal online activities

Capstone Projects

Project 1: School Network Design

Integration: Science + ICT

Description: Students design a network layout for a hypothetical school campus, identifying devices, connections, and resource sharing requirements. They create network diagrams and present their proposals.

Skills: Network topologies, LAN/WAN concepts, diagramming, presentation

Project 2: Community Data Dashboard

Integration: Mathematics + ICT

Description: Students collect data about their community (population, rainfall, agricultural produce) and create a spreadsheet with charts and formulas to visualize and analyze the information.

Skills: Spreadsheet formulas, chart creation, data analysis, digital presentation